Lecture 7 - Local Average Treatment Effects

Alice Wu

UW Madison Econ 695

Recap

Remember + Put on cheatsheet

- Continue our discussion of causal effects under the potential outcomes framework $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
- We have shown that the average treatment effect $E[\beta_i] = E[Y_i(1) - Y_i(0)]$ can be point identified in the following cases:
  - Randomized control trials, where $(Y_i(0), Y_i(1)) \perp D_i$
  - propensity score → Selection on observables: $(Y_i(0), Y_i(1)) \perp D_i | X_i$ (can estimate CATE first or estimate propensity scores, and form $\frac{D_i Y_i}{\hat{p}_i}$, $\frac{(1-D_i)Y_i}{1-\hat{p}_i}$
  - no complete perfect compliance → IV method if RCT has incomplete compliance: constant treatment effect $\beta_1 \equiv Y_i(1) - Y_i(0)$ can be identified via an instrument Z_i for D_i:

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}^1 - \bar{Y}^0}{\bar{D}^1 - \bar{D}^0} \xrightarrow{p} \beta_1 \text{ constant TE} \sim \text{ATE}$$

where $\bar{Y}^1 = \frac{1}{N_1} \sum_{i:z_i=1} y_i$ (sample mean among those randomly assigned to $Z_i = 1$).

This Lecture

- This lecture we continue to work with the general framework allowing for treatment effect heterogeneity:

$$\beta_{1i} := Y_i(1) - Y_i(0)$$

- Can we identify ATE $E[\beta_{1i}]$ using the IV method, or the ATE for some sub-population? What does $\hat{\beta}_1^{IV}$ converge to?
- Local average treatment effects (IV-LATE)
  - ATE for compliers (but not always takers/never takers/defiers)
  - characterize compliers, in comparison with other groups.
  - Application: Self-Sufficiency Project, C-section

people who remove kids when head start starts

Heterogeneous Treatment Effects

- Causal model of interest:

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$$\begin{aligned} Y_i &= D_i Y_i(1) + (1 - D_i) Y_i(0) \\ &= Y_i(0) + D_i \times (Y_i(1) - Y_i(0)) \\ &= Y_i(0) + D_i \times \boldsymbol{\beta_{1i}} \end{aligned} \tag{1}$$

- Define ATE $\beta_1 = E[\beta_{1i}]$, and $\beta_0 := E[Y_i(0)]$, (1) can be rewritten as:

$= Y_i(0) + \beta_{1i} \cdot D_i = \beta_0 + (Y_i(0) - E[Y_i(0)]) + \beta_1 D_i + (\beta_{1i} - \beta_1) D_i$

D_i u_i could be correlated

$$Y_i = \beta_0 + \beta_1 D_i + u_i$$
$$\text{where } u_i = Y_i(0) - \beta_0 + (\beta_{1i} - \beta_1) D_i$$

the residual u_i is clearly correlated with D_i unless $D_i \perp (Y_i(0), Y_i(1))$ (as in a perfect RCT).

- If you estimate a population regression $Y_i = b_0 + b_1 D_i + \epsilon_i$, what is b_1?

Instrument Variable Approach

- Suppose we have a valid instrument for treatment D_i: Z_i predicts D_i and $E[Z_i u_i] = 0$. Turns out the IV approach can help us get consistent estimates for the average treatment effects in some **sub-population (compliers)**.
- Let's look at an *experiment to study the effect of smoking by mothers on birthweight.*
    - Why not a RCT? Can't command mothers-to-be to smoke or not. So: conduct an "encouragement design" experiment among pregnant women who smoke and want to quit.
    - Z_i = **treatment** status = 1 if given "treatment" (counseling)
    - D_i =dummy for "desist" or stop smoking (1=desist, 0=keep smoking)

Instrument Variable Approach

- Y_i =weight of newly born infant

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 D_i + u_i \quad \text{(causal model)} \qquad (2) \\ D_i &= \pi_0 + \pi_1 Z_i + \eta_i \quad \text{(1st stage)} \\ Y_i &= \underbrace{\beta_0 + \beta_1 \pi_0}_{\delta_0} + \underbrace{\beta_1 \pi_1}_{\delta_1} Z_i + \underbrace{(u_i + \beta_1 \eta_i)}_{\nu_i} \end{aligned}$$

by exclusion restriction and the first-stage regression, $Z_i \perp (u_i, \eta_i)$. Therefore, we can consistently estimate the reduced-form regression:

$$Y_i = \delta_0 + \delta_1 Z_i + \nu_i$$
$$\delta_1 = \beta_1 \pi_1$$

- This is the "dummy-dummy" case (both D_i and Z_i are binary). What are first stage and reduced form coefficients in sample?

$$\begin{aligned} \hat{\pi}_1 &= \overline{D}^1 - \overline{D}^0 \\ \hat{\delta}_1 &= \overline{Y}^1 - \overline{Y}^0 \\ \Rightarrow \quad \hat{\beta}_1^{iv} &= \frac{\overline{Y}^1 - \overline{Y}^0}{\overline{D}^1 - \overline{D}^0} \qquad (3) \end{aligned}$$

Instrument Variable Approach

- We are dividing the difference in birthweights (between treatment and control groups) by the difference in the fraction of mothers who quit.

- What does $\hat{\beta}_1^{iv}$ represent? From (3), note $\hat{\beta}_1^{iv}$ is driven by the gains in birth outcomes for the **moms who quit smoking because of the encouragement**.
- Let's formalize this interpretation!

Instrument Variable in Potential Outcome Framework

- Let's think about how people behave and how they can respond to "treatment" ($Z_i = 1$, e.g. receive counseling).
- Let's introduce 2 new random variables in the potential outcome framework – that describe types of people and how they react: $(D_i(0), D_i(1))$
  - $D_i(0)$ = indicator of desist if person i is assigned to $z = 0$
  - $D_i(1)$ = indicator of desist if person i is assigned to $z = 1$
- Now we can characterize/classify individuals based on their **compliance type** $(D_i(0), D_i(1))$:
  - $D_i(0) = 0$, $D_i(1) = 0$ – "never takers" - never desist
  - $D_i(0) = 1$, $D_i(1) = 1$ – "always takers" - always desist
  - $D_i(0) = 0$, $D_i(1) = 1$ – "compliers" - desist if treated
  - $D_i(0) = 1$, $D_i(1) = 0$ – "defiers" - desist if untreated, not if treated.
- **(Random assignment)** All 4 groups should be equally likely in T and C groups.

$$(D_i(0), D_i(1)) \perp Z_i$$

which implies $\Rightarrow Pr(\text{Compliers}|Z_i) = Pr(\text{Compliers})$, and also for AT/NT/Defiers.

IV - First Stage

- Who is still smoking when kid is born?
  - in Control-group ($Z_i = 0$) : smokers=$NT's + Comp's$
  - in Treated-group ($Z_i = 1$) : smokers= $NT's$ (if no defiers)
- So: if no defiers, difference in fraction of smokers = fraction of **Compliers**! The first-stage under the random assignment assumption:

$$
\begin{aligned}
D_i &= \pi_0 + \pi_1 Z_i + \eta_i \\
\pi_1 &= E[D_i|Z_i = 1] - E[D_i|Z_i = 0] \\
&= E[D_i(1)|Z_i = 1] - E[D_i(0)|Z_i = 0] \\
&= Pr(D_i(1) = 1|Z_i = 1) - Pr(D_i(0) = 1|Z_i = 0) \\
&= Pr(D_i(1) = 1) - Pr(D_i(0) = 1) \text{ by random assignment} \\
&= Pr(AT + Comp) - Pr(AT + Defiers)
\end{aligned}
$$

The assumption $Pr(Defiers) = 0$ is called the **monotonicity assumption**:

$$\forall i : \; D_i(1) \geq D_i(0)$$

IV - First Stage

  - note: if we flip the definition of D_i = smoking, this assumption is $\forall i : D_i(1) \leq D_i(0)$. There is no defier with $D_i(1) = 1$ while $D_i(0) = 0$.

- In summary, under the random assignment + monotonicity assumptions:

$$\pi_1 = Pr(Comp)$$

  the first-stage coefficient is the fraction of compliers (characterized by $D_i(1) = 1, D_i(0) = 0$) in the population.

IV - 2nd Stage $Y_i(d, z)$

- Clearly the difference in birth weights between the $z = 0$ and $z = 1$ groups depends on the changes experienced by the compliers as a result of the intervention in the $z = 1$ environment. To formalize that we revise the *potential outcomes* for y:

  $$Y_i(d, z) = \text{outcome for } i \text{ if } D_i = d \textit{ and } Z_i = z$$

  Now if z has no direct effect on the outcome:

  $$\begin{aligned} Y_i(0,0) &= Y_i(0,1) = Y_i(0) \text{ the outcome if } D_i = 0 \\ Y_i(1,0) &= Y_i(1,1) = Y_i(1) \text{ the outcome if } D_i = 1 \end{aligned}$$

  where $Y_i(d)$ is the potential outcome when unit i has $D_i = d$ (as before).

- We often write the **exclusion restriction** assumption as:

  $$Y_i(d, z) = Y_i(d)$$

IV - 2nd Stage $Y_i(d, z)$

- Recap: Part of what it means to be a "good instrument" is that - z only works through its effect on D. We will assume this "exclusion restriction" is true. When we write

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 D_i + u_i \quad \text{(causal model)} \\ D_i &= \pi_0 + \pi_1 Z_i + \eta_i \quad \text{(1st stage)} \end{aligned}$$

  we are *building in the exclusion restriction*. Note that
  - Z_i does not appear in the structural model
  - since $E[Z_i u_i] = 0$, changes in z_i do not affect u_i. Thus the ONLY way for Z_i to affect Y_i is via the effect on X_i.

IV - 2nd Stage: Realized Y_i by group

The realized outcome for unit i can be expressed as:

$$Y_i = Y_i(D_i) = Y_i(D_i(Z_i)) \tag{4}$$

Let's break it down by group:

- Never takers ($D_i(0) = 0, D_i(1) = 0$): $D_i = 0$ regardless of $z_i \Rightarrow$ $Y_i = Y_i(0)$ for either z_i.
- Compliers ($D_i(0) = 0, D_i(1) = 1$): observe $Y_i = Y_i(0)$ if $z_i = 0$, observe $Y_i = Y_i(1)$ if $z_i = 1$
- Always takers ($D_i(0) = 1, D_i(1) = 1$): $D_i = 1$ regardless of $z_i \Rightarrow$ $Y_i = Y_i(1)$ for either z_i
- Defiers ($D_i(0) = 1, D_i(1) = 0$): observe $Y_i = Y_i(1)$ if $z_i = 0$, $D_i = 0$ if $z_i = 1 \Rightarrow$ observe $Y_i = Y_i(0)$ if $z_i = 1$

IV - 2nd Stage: Realized Y_i by group

Now let's think about the mean outcomes we *observe* in the $z = 0$ world:

$$\begin{aligned}
E[Y_i|Z_i = 0] &= E[Y_i(D_i(0))\,|\,Z_i = 0] \text{ by exclusion restriction} \\
&= E_g[E[Y_i(D_i(0))\,|Z_i = 0, g]] \text{ by the law of iterated exp.} \\
&= \sum_{g\in\{NT,AT,Comp,Def\}} E[Y_i(D_i(0))\,|g, Z_i = 0] \times Pr(g|Z_i = 0) \\
&= E[Y_i(0)|NT] \times Pr(NT) \\
&\quad +E[Y_i(0)|Comp] \times Pr(Comp) \\
&\quad +E[Y_i(1)|AT] \times Pr(AT) \\
&\quad +E[Y_i(1)|Def] \times Pr(Def)
\end{aligned}$$

the last equation comes from the random assignment assumption $(D_i(1), D_i(0)) \perp Z_i$. So fractions of the various groups ($e.g.$, $Pr(NT)$,etc) and expected potential outcomes (e.g., $E[Y_i(1)|AT]$,etc) do not depend on z=0 or 1 status.

IV - 2nd Stage: Realized Y_i by group

Likewise in the $z = 1$ world:

$$\begin{aligned} E[Y_i|Z_i = 1] &= E[Y_i(0)|NT] \times Pr(NT) \\ &+E[Y_i(1)|Comp] \times Pr(Comp) \\ &+E[Y_i(1)|AT] \times Pr(AT) \\ &+E[Y_i(0)|Def] \times Pr(Def) \end{aligned}$$

If there are no defiers (under monotonicity assumption $D_i(1) \geq D_i(0)$):

$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[Y_{i1} - Y_{i0}|Comp] \times Pr(Comp)$$

IV - 2nd Stage: Realized Y_i by group

Thus the population regression version of the reduced form model yields

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp)$$

and the population regression version of the 1st stage model yields

$$\pi_1 = Pr(Comp) = E[D_i|Z_i = 1] - E[D_i|Z_i = 0]$$

So

$$\frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0) \mid Comp]$$

Under our assumptions, the ratio of the reduced form effect δ_1 to the 1st stage effect π_1 is the average treatment effect for compliers, which is called the local average treatment effects (local in the sense that $D_i(1) > D_i(0)$).

IV-LATE: Summary

Let's summarize the assumptions we have made in a IV-LATE framework, where we can classify individuals into four groups based on

$$(D_i(0), D_i(1)) = \begin{cases} (0,0) & \text{never takers} \\ (0,1) & \text{compliers} \\ (1,0) & \text{defiers} \\ (1,1) & \text{always takers} \end{cases}$$

1. **Random assignment:** treatment Z_i is randomly assigned. All four groups are equally likely to be in treated or in control.

$$(D_i(0), D_i(1)) \perp Z_i$$

2. **Exclusion Restriction:** (instrument) Z_i matters for outcome Y_i only through D_i

$$Y_i(d, z) = Y_i(d)$$

   - Note 1 & 2 implies the joint distribution $(D_i(0), D_i(1), Y_i(0), Y_i(1))$ is the same whether $Z_i = 0$ or $Z_i = 1$.

3. **Monotonicity:** $D_i(1) \geq D_i(0)$ for all units. In other words, there are no defiers.

IV-LATE: Summary

Under the assumptions above, the first-stage regression gives us:

$$\begin{aligned} D_i &= \pi_0 + \pi_1 Z_i + \eta_i \\ \pi_1 &= E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = Pr(Comp) \end{aligned}$$

and the reduced-form regression:

$$\begin{aligned} Y_i &= \delta_0 + \delta_1 Z_i + \nu_i \\ \delta_1 &= E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] \\ &= E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp) \end{aligned}$$

therefore the ratio, which is the population "IV":

$$\beta_1^{IV} = \frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0)|Comp]$$

The IV estimate provides an estimate of the "gain" in Y that compliers obtain as a result of being in the $z = 1$ setting and getting $D_i = 1$ instead of $D_i = 0$.

LATE vs. ATE

- Take a closer look at the original model (heterogeneous treatment effects):
$$Y_i = \beta_{0i} + \beta_{1i} D_i = \beta_0 + \beta_1 D_i + u_i$$
where $\beta_1 = E[Y_i(1) - Y_i(0)] = E[\beta_{1i}]$.
- IV estimator:
$$\hat{\beta}_1^{IV} \xrightarrow{p} E[Y_i(1) - Y_i(0)|Comp] \text{ LATE}$$
which is different from ATE $\beta_1$for the entire population.
- When does LATE = ATE?
  1. (Boring case) Constant treatment effect: $Y_i(1) - Y_i(0) \equiv \beta_1$, we showed in the previous lecture that IV yields a consistent estimate of the constant TE.
  2. Homogeneous TE across sub-populations:
$$E[Y_i(1) - Y_i(0)|AT] = E[Y_i(1) - Y_i(0)|NT] = E[Y_i(1) - Y_i(0)|Comp]$$
$$\Rightarrow E[Y_i(1) - Y_i(0)|Comp] = E[Y_i(1) - Y_i(0)]$$

Extrapolation

- Is it plausible $E[Y_i(1) - Y_i(0)|Comp] = E[Y_i(1) - Y_i(0)|AT] = E[Y_i(1) - Y_i(0)|NT]$?
- From the data, we can estimate $E[Y_i(0)|NT]$ and $E[Y_i(1)|AT]$:

$$E[Y_i|D_i = 0, Z_i = 1] = E[Y_i(0)|NT, Z = 1] = E[Y_i(0)|NT]$$
$$E[Y_i|D_i = 1, Z_i = 0] = E[Y_i(1)|AT, Z = 0] = E[Y_i(1)|AT]$$

and also $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$:

$$\begin{aligned} E[Y_i|D_i = 0, Z_i = 0] =& \frac{Pr(NT)}{Pr(NT) + Pr(Comp)} \times E[Y_i(0)|NT] + \\ &+ \frac{Pr(Comp)}{Pr(NT) + Pr(Comp)} \times \boldsymbol{E[Y_i(0)|Comp]} \end{aligned}$$

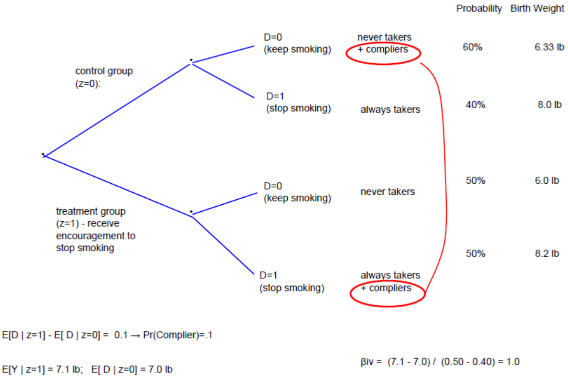
Question: how do you estimate $Pr(NT)$ or $Pr(AT)$?

- Comparing $E[Y_i(0)|NT]$ with $E[Y_i(0)|Comp]$, and $E[Y_i(1)|AT]$ with $E[Y_i(1)|Comp]$, if they are substantially different in levels, then it is much less plausible the ATE for compliers is similar to the average effects for other compliance types.

Example: Smoking

Let's revisit the *experiment to study the effect of smoking by mothers on birthweight* (Y_i). $Z_i = 1$ if given "treatment" (counseling). D_i =dummy for "desist" or stop smoking (1=desist, 0=keep smoking)

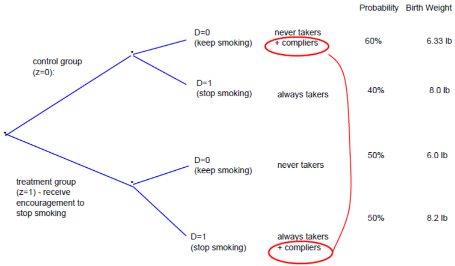
- First stage: $\hat{\pi}_1 = \bar{D}^1 - \bar{D}^0 = 0.5 - 0.4 = Pr(Comp)$
- Second stage: $\hat{\delta}_1 = \bar{Y}^1 - \bar{Y}^0 = (6 * 0.5 + 8.2 * 0.5) - (6.33 * 0.6 + 0.40 * 8) \approx 0.10$



Example: Smoking

Is it plausible the ATE for AT or NT are similar to LATE?

- $E[Y_i(1)|AT]$ is approximated by the mean when $D = 0, Z = 1$: 8 lbs
- $E[Y_i|Z_i = 1, D_i = 1] = \frac{Pr(AT)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|AT] + \frac{Pr(Comp)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|Comp] \Rightarrow$: $(8.2 - \frac{0.4}{0.5} * 8)/0.2 = 9$ is a consistent estimate of $E[Y_i(1)|Comp]$
- $E[Y_i(0)|Comp] = E[Y_i(1)|Comp] - \beta_1^{IV}$: about 8 lbs



E[D | z=1] - E[D | z=0] = 0.1 → Pr(Complier)=.1

E[Y | z=1] = 7.1 lb; E[D | z=0] = 7.0 lb

βiv = (7.1 - 7.0) / (0.50 - 0.40) = 1.0

Characterize Compliers vs. AT, NT

- Let X_i denote some characteristics for unit i. Another way to see if make sense to extrapolate LATE to other compliance types is to compare observable characteristics of compliers vs. others. It's helpful to know how compliers as a subpopulation are compared with the overall population.
- We can estimate the following from the data:

$$
\begin{aligned}
E[X_i|D_i = 0, Z_i = 1] =& E[X_i|NT] \\
E[X_i|D_i = 0, Z_i = 0] =& E[X_i|Comp] \times \frac{Pr(Comp)}{Pr(Comp) + Pr(NT)} \\
& + E[X_i|NT] \times \frac{Pr(NT)}{Pr(Comp) + Pr(NT)} \\
E[X_i|D_i = 1, Z_i = 0] =& E[X_i|AT] \\
E[X_i|D_i = 1, Z_i = 1] =& ...
\end{aligned}
$$

You could calculate the mean of X by (D, Z) and work out $\bar{X}^{Comp}$.

Characterize Compliers vs. AT, NT

- But there is an easier way! Define outcome $Y_i = X_i D_i$. Consider this goofy 2SLS model:

$$\begin{aligned} \text{Causal Model: } X_i D_i &= a + bD_i + e_i \\ \text{First stage: } D_i &= \pi_0 + \pi_1 Z_i + \eta_i \end{aligned} \tag{5}$$

which yields a reduced-form regression of $X_i D_i$ on instrument Z_i

$$\begin{aligned} X_i D_i =& \delta_0 + \delta_1 Z_i + \nu_i \\ \delta_1 =& E[X_i D_i | Z_i = 1] - E[X_i D_i | Z_i = 0] \\ =& E_g[E[X_i D_i(1) | Z_i = 1, g]] - E_g[E[X_i D_i(0) | Z_i = 0, g]] \end{aligned}$$

Note $X_i D_i(1) = X_i$ if $D_i(1) = 1$, 0 otherwise, and $X_i D_i(0) = X_i$ for always takers. Show:

$$\begin{aligned} \delta_1 &= E[X_i | Comp] \times Pr(Comp) \\ b^{IV} &= \frac{\delta_1}{\pi_1} = E[X | Comp] \end{aligned}$$

we have got the mean characteristics of compliers!

$(Y_i(0), Y_i(1))$ as Characteristics

- Potential outcomes of each unit i can be thought of as characteristics X_i. We can apply/adjust the goofy model above to estimate $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$.
- Note $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) \sim$ we observe $Y_i(1)$ among units with $D_i = 1$, and $Y_i(0)$ among $D_i = 0$.
- First, construct outcome

$$Y_i(1) \times D_i = \begin{cases} Y_i & D_i = 1 \\ 0 & D_i = 0 \end{cases}$$

and consider "goofy" 2SLS:

$$\begin{aligned} \text{Causal Model: } Y_i(1) D_i &= a_1 + b_1 D_i + {}_i \\ \text{First stage: } D_i &= \pi_0 + \pi_1 Z_i + \eta_i \end{aligned} \tag{6}$$

$(Y_i(0), Y_i(1))$ as Characteristics

Reduced-form $Y_i(1)D_i = \delta_0 + \delta_1 Z_i + \nu_i$ can be consistently estimated, in which

$$\begin{aligned} \delta_1 &= E[Y_i(1)D_i|Z_i = 1] - E[Y_i(1)D_i|Z_i = 0] \\ &= E[Y_i(1)|Comp] \times Pr(Comp) \\ b_1^{IV} &= E[Y_i(1)|Comp] \end{aligned}$$

$(Y_i(0), Y_i(1))$ as Characteristics

- Second, construct outcome

$$Y_i(0) \times (1 - D_i) = \begin{cases} 0 & D_i = 1 \\ Y_i(0) & D_i = 0 \end{cases}$$

and revise the goofy 2SLS:

$$\begin{aligned} \text{Causal Model: } Y_i(0)(1 - D_i) &= a_0 + b_0 D_i + \xi_i \\ \text{First stage: } D_i &= \pi_0 + \pi_1 Z_i + \eta_i \end{aligned} \tag{7}$$

the reduced-form coef. on Z_i becomes:

$$\begin{aligned} \delta_1 &= E[Y_i(0)(1 - D_i)|Z_i = 1] - E[Y_i(0)(1 - D_i)|Z_i = 0] \\ &= -E[Y_i(0)|Comp] \times Pr(Comp) \\ b_0^{IV} &= -E[Y_i(0)|Comp] \end{aligned}$$

- The IV estimates b_1^{IV}, b_0^{IV} are linked to LATE:

$$E[Y_i(1) - Y_i(0)|Comp] = b_1^{IV} + b_0^{IV}$$

Is this surprising? $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$.

Application: Self-Sufficiency Program

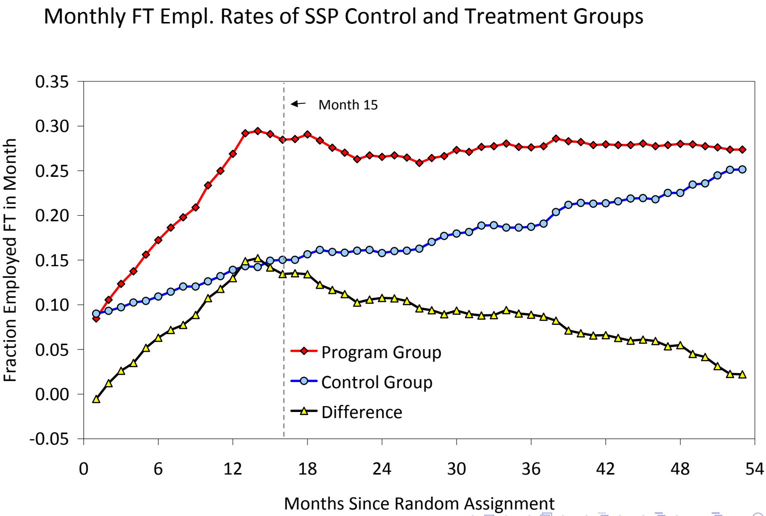
Example from Card and Hyslop (2005):

- Self Sufficiency Project (SSP) experiment, conducted in Canada, early 1990s
- Earnings subsidy (negative tax) for people who enter full-time (FT) work
- Subsidy $= 0.5 \times$(T-monthly earnings) T=\$2500/mo or \$3083/mo
- ONLY for the FT (30+ hrs/week) who earn $\geq$ the minimum wage:
  - subsidy lasts for 3 years;
  - have to start receiving subsidy within 12 mo. of random assignment.
- ***Research Questions***: SSP subsidy = strong incentive to enter FT work, which is a "short-term establishment incentive".
  1. can long-term welfare recipients find a job and leave welfare? (short-term establishment)
  2. can they maintain a FT job and be better off in the long-term? (long-term entitlement as they choose work over welfare)

Experiment:

- Welfare recipients with kids (95% females), 1+ years on welfare
- Randomly allocated to treatment group ($z = 1$) or control ($z = 0$)
- Series of surveys measure employment etc.

Application: Self-Sufficiency Program



Causal Model

- Outcome "Y": $IA15_i = 1$ if on welfare in month 15
- Endogenous variable of interest "D": $FT15_i = 1$ if the worker is employed full-time in month 15.
- Instrument "Z": random assignment to treatment group in the SSP experiment
- Causal model for receiving income assistance (IA) at month 15:

$$IA15_i = \beta_0 + \beta_1 FT15_i + u_i$$

- First stage model for FT empl at month 15:

$$FT15_i = \pi_0 + \pi_1 TREAT_i + \eta_i$$

- Reduced form model for IA at month 15:

$$IA15_i = \delta_0 + \delta_1 TREAT_i + \nu_i$$

Causal Model

- Underlying assumptions:
  1. Treatment is randomly assigned
     $$(FT(0), FT(1)) \perp TREAT_i$$
  2. Exclusion restriction $Y(d,z) = Y(d)$: SSP only affects IA through full-time work? $SSP\ subsidy \Rightarrow FT\ work \Rightarrow leave\ IA$ seems plausible.
  3. Is "no defiers" satisfied? Likely since SSP would never cause you to stop working FT (unless you are really weird).

Estimate LATE

Summary of SSP Outcomes at Month 15

	Employed FT (1)	On Welfare (IA) (2)
Control Group (n=2718)	0.1472	0.8098
Treatment Group (n=2762)	0.2848	0.6687
Difference: T-C (std error)	0.1376 (0.011)	-0.1411 (0.012)
IV estimate:	-1.027 (0.082)	

Estimate LATE

We know that there is only partial compliance with the subsidy incentive.
$Pr(NT,\ mo.15) = 1 - 0.2848 = 0.7152$
$Pr(AT,\ mo.15) =$0.1472
$Pr(C,\ mo.15) = 0.1376$
Note that these fractions are changing as the experiment goes on (x-axis = months since random assignment) ....
Q: who are the compliers? Are the "like" other people? Are they more or less advantaged?

Characterize Compliers

- X_i = HS graduation
- Apply goofy 2SLS to outcome "D*X" ~ $X_i \times FT15_i$

$$X_i\, FT15_i = a + b \times FT15_i + u_i$$
$$\widehat{FT15}_i = \pi_0 + \pi_1 TREAT_i$$

$$b^{IV} = \frac{cov(X_i FT15_i,\ \widehat{FT15}_i)}{var(\widehat{FT15})} = \frac{1}{\pi_1} \frac{cov(X_i FT15_i, Z_i)}{var(Z_i)}$$
$$= \frac{E[X_i FT15_i | Z_i = 1] - E[X_i FT15_i | Z_i = 0]}{E[FT15_i | Z_i = 1] - E[FT15_i | Z_i = 0]}$$

Characterize Compliers

Two-Stage Least Squares Estimation

Model	d_hsgrad
Dependent Variable	d_hsgrad

Analysis of VarianceS

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	7.401383	7.401383	139.92	<.0001
Error	5478	289.7814	0.052899		
Corrected Total	5479	609.8392			

Root MSE	0.23000	R-Square	0.02491
Dependent Mean	0.12755	Adj R-Sq	0.02473
Coeff Var	180.31324		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variable Label
Intercept	1	0.011784	0.010269	1.15	0.2512	Intercept
EMFTM15	1	0.534926	0.045223	11.83	<.0001	Msurv:Employed full-time in mo

Application: The Health Impacts of C-section

- Card, Fenizia, Silver: "The Health Impacts of Hospital Delivery Practices"
- Research Question: treatment practices vary widely across hospitals. In the context of childbirth, the authors assess impacts of the differences in delivery practices at childbirth on infant and maternal health
- Data: births in CA, with detailed information such as where patients live, where they deliver kids, delivery practices, health outcomes through records of ER visits and medical claims...
  - they classify hospitals based on c-section rates: H if a "high c-section" hospital, L otherwise
  - D_i = deliver at H hospital
  - y_i = c-section delivery; baby is readmitted to hospital after birth
- Empirical challenge: there is endogenous sorting in c-section (e.g., high risk, unobserved preferences...)
- Quasi-experiment Design:
  - living closer to a H hospital than a L *exogenously* increases the chance that they choose a H hospital;

Application: The Health Impacts of C-section

- Z_i= relative distance of mom's hom to H vs L hospital (continuous or indicator)
- There are 2 types of compliers:
    - H-compliers: mothers who choose a H hospital because they live closer to H;
    - $H\&C$-compliers: mothers who choose a H hospital because they live closer to H and will choose c-section; if they live closer to a L they would not have used c-section.

First-stage: "H" or "H+C" Compliers

Assume: (1) random assign (2) exclusion restriction and (3) monotonicity.

$$\text{deliver at H hospital:} H_i = \pi_0 + \pi_1 Z_i + \pi_X X_i + \epsilon_i$$
$$\text{deliver by C-section:} C_i = \gamma_0 + \gamma_1 Z_i + \gamma_X X_i + \eta_i$$

Interpretations:

- π_1 : the fraction of H-compliers, who will choose a H hospital if they live closer to H. Note this includes people who would have chosen C-section at both (H, L) hospitals ("C" always takers), and people who were to choose c-section if they chose a H-hospital ("H+C" compliers).
- γ_1 : the fraction of "H+C" compliers. Why? Under monotonicity assumption, the only group who changes C_i are $H + C$ compliers who choose a H hospital because they live closer to H, and who choose C-section because they are at H.

First-stage: "H" or "H+C" Compliers

Table III: Estimated Effects of Relative Distance on Place, Mode, and Timing of Delivery

		Instrument=Relative Distance to H Hospital Coefficients × 100		Instrument= Indicator for Closer to H Hospital Coefficients × 10	
	Mean	Baseline controls only	All controls	Baseline controls only	All controls
Outcome Variable	(1)	(2)	(3)	(4)	(5)
Deliver at H Hospital	0.515	1.304	1.328	0.682	0.694
		(0.135)	(0.134)	(0.094)	(0.091)
C-section Delivery	0.256	0.159	0.149	0.078	0.080
		(0.028)	(0.027)	(0.021)	(0.020)
Breakdown of C-Section Deliveries:					
C-Section at H Hospital	0.149	0.401	0.403	0.208	0.213
		(0.041)	(0.041)	(0.030)	(0.030)
C-Section at L Hospital	0.106	-0.243	-0.254	-0.130	-0.133
		(0.033)	(0.030)	(0.024)	(0.022)
Fractions of Complier Groups -- Moving 7 mi. closer to H hospital (col. 2-3) or closer to H hospital (col. 4-5)					
P(H Complier)		0.091	0.093	0.068	0.069
P(C&H Complier)		0.011	0.010	0.008	0.008
P(H Complier & C Always-Taker)		0.017	0.018	0.013	0.013

IV-LATE for "H" Compliers

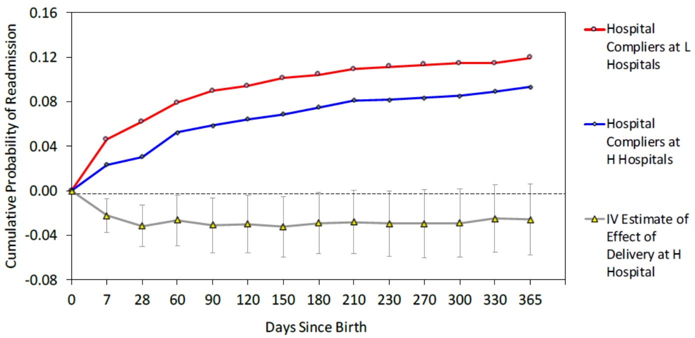
Focusing on the H-hospital compliers, estimate the 2SLS model:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 H_i + \beta_X X_i + u_i \\ &= \delta_0 + \delta_1 Z_i + \delta_X X_i + \nu_i \\ \Rightarrow \beta_1^{IV} &= \frac{\delta_1}{\pi_1} \end{aligned}$$

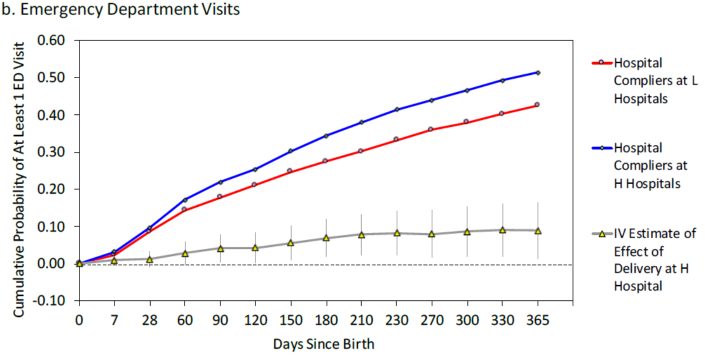
the IV estimate represents the LATE for H-compliers – the change in outcome y per additional delivery due to *H* hospital among the compliers. The authors find mixed health impacts: in the short run readmission rate is lower, but in the longer run ER visits are higher among the compliers.

IV-LATE for "H" Compliers

a. Hospital Readmissions



IV-LATE for “H” Compliers



Question: How did they estimate $E[Y_i(1)|Comp]$, $E[Y_i(0)|Comp]$? Hint: goofy 2sls on $Y_i D_i$ and $Y_i(1 - D_i)$

Characteristics of Compliers

- The "relative distance" instrument is continuous. But we can apply the goofy 2SLS to characterize the compliers:

$$\text{Causal Model: } X_i D_i = a + bD_i + e_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$b^{IV} = \frac{cov(X_i D_i, Z_i)}{cov(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i])\,|D = 1]}{E[Z_i - E[Z_i]\,|D = 1]}$$

Table IV: Characteristics of Compliers

	Analysis Sample	Compliers		Ratio: C's/All (%)
Socio-Economic Characteristics of Mother				
Race/Ethnicity				
White	31.7%	25.9	(5.1)	81.6
Black	5.6%	0.7	(1.5)	12.4
Asian	17.6%	11.3	(4.8)	64.1
Hispanic	44.2%	62.2	(5.7)	140.7
Education				
High School or Less	41.2%	54.4	(4.7)	132.0
Some College	20.1%	12.9	(2.1)	64.5
BA or Higher	38.7%	32.6	(5.3)	84.4
Government Insurance	43.3%	62.4	(5.7)	144.2
Predicted Probabilities of Infant ED Visit/Inpatient Admissions				
Prob(ED Visit) > median	50.0%	67.8	(6.3)	135.5
Prob(Inpat. Adm) > median	50.0%	71.0	(6.4)	142.1

Characterize Compliers - Continuous Instrument

$$b^{IV} = \frac{cov(X_i D_i, Z_i)}{cov(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i])\,|D=1]}{E[Z_i - E[Z_i]\,|D=1]}$$
$$= \int_{\underline{z}} E[X|Comp(\underline{z})] \times \omega(\underline{z}) d\underline{z}$$

is weighted average of $E[X|Comp(\underline{z})]$ across complier groups, with weights $\int \omega(\underline{z}) d\underline{z} = 1$. Think of $Comp(\underline{z})$ as compliers who would have $D_i(z) = 1[Z_i \geq \underline{z}]$. For always takers, $\underline{z} = -\infty$, and for never takers, $\underline{z} = \infty$. In the binary instrument case, there is only 1 group of compliers: $D_i(1) > D_i(0)$. (Optional): Prove $b^{IV} = \int_{\underline{z}} E[X|Comp(\underline{z})] \times \omega(\underline{z}) d\underline{z}$ for some weights $\int \omega(\underline{z}) d\underline{z} = 1$.

- Classify units by $\underline{z}$ where $D_i(z) = 1[Z_i \geq \underline{z}]$
- Denominator (applying random assign of Z):

$$E[\widetilde{Z}|D=1] = \int_{\underline{z}} E[\widetilde{Z}|\underbrace{Z \geq \underline{z}}_{D=1}, \underline{z}] dF(\underline{z}) = \int_{\underline{z}} E[\widetilde{Z}|Z \geq \underline{z}] dF(\underline{z})$$

Characterize Compliers - Continuous Instrument

- Numerator:

$$E[X\widetilde{Z}|D=1] = \int_{\underline{z}} E[X\widetilde{Z}|Z \geq \underline{z}, \underline{z}]dF(\underline{z}) = \int_{\underline{z}} E[X|\underline{z}] \times E[\widetilde{Z}|Z \geq \underline{z}]dF(\underline{z})$$

- So the ratio

$$\frac{E[X\widetilde{Z}|D=1]}{E[\widetilde{Z}|D=1]} = \int_{\underline{z}} E[X|\underline{z}] \times \underbrace{\frac{E[\widetilde{Z}|Z \geq \underline{z}]f(\underline{z})}{\int_{\underline{z}} E[\widetilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}}}_{\text{weight } \omega(\underline{z})} d\underline{z}$$